

Revision Memorandum

Quantum Entanglement as Boundary-Level Compatibility

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Overview

This revision upgrades the manuscript into the Bell-witness member of the paired AASC quantum determinacy arc and makes no-signaling the flagship endpoint. It develops the positive determinacy structure behind correlation without communication: AMetric co-standing and boundary co-fixation. The final precision pass clarifies the statistical status of Bell correlation, the type of the tensor-compatibility content T_B , and the priority sense of “before” at the AMetric boundary. The manuscript remains faithful to the original skeleton: Bell scope, imported admissibility substrate, Bell witness fixation, boundary traces, joint standing classes, local measurement as realization, no hidden variables, no signaling, AMetric boundary, CHSH audit, and construction-space closure.

Major changes

1. Retitled the manuscript as a kernel-role determinacy theorem for no-signaling, Bell witness fixation, and tensor-level compatibility.
2. Front-loaded no-signaling as the flagship same-domain determinacy endpoint rather than a merely preserved background condition.
3. Added the dedicated section “AMetric co-standing and boundary co-fixation.”
4. Defined the co-standing vocabulary: local Bell face, boundary co-fixation, T_B , and admissible realization pair.
5. Added the theorem “AMetric co-standing yields correlation without communication.”
6. Clarified that T_B fixes the admissible joint realization profile and associated operational probabilities, not deterministic pre-assigned local values.
7. Clarified that individual outcome pairs are local realizations, while the family of such pairs carries the Bell correlation structure.
8. Added the explicit determinacy chain: entangled preparation $\Rightarrow$ joint standing class $\Rightarrow$ AMetric co-standing $\Rightarrow$ tensor compatibility $\Rightarrow$ realization profile $\Rightarrow$ Bell correlation without communication.
9. Added a communication-reading versus AMetric-co-standing table.
10. Strengthened the no-signaling endpoint proof by routing correlation through AMetric co-standing before excluding signal, transport, marginal-modulation, and distant-repair typings.

11. Added DOI-backed support references for the AMetric boundary and Boundary-Trace Fixation papers.
12. Updated the conclusion and appendix certificate so no-signaling is explicitly “correlation without communication” through AMetric co-standing.
13. Added the admissibility-priority clarification: “before” at the AMetric boundary is not temporal priority.

Support-source integration

The AMetric boundary source is used to support the claim that boundary-side fixation admits no metric, ordering, ranking, probability, selector, canonical representative, or transport authority at admission. The Boundary-Trace source is used to support the declared, stable, bivalent, channel-preserving trace discipline that fixes Bell witness interfaces without environmental oracle use or post-selection repair.

Build and QA

The manuscript builds successfully under `pdflatex`. The final PDF is 32 pages. Final build logs report no undefined references, undefined citations, or overfull-box warnings. The PDF was rendered to page images for visual QA, and the final package integrity check passed.